Phone: +1-805-796-4754
Email: Kadenlee84@gmail.com
Portfolio: klee-eng.github.io

Kaden Lee

Education

University of California, San Diego *09/2025 - 06/2026*
MS in Mechatronics and Controls GPA: 3.82/4.0

University of California, Santa Barbara *10/2020 - 06/2024*
BS in Mechanical Engineering GPA: 3.74/4.0

Relevant Coursework: *Robotics, Robot Motion Planning and Learning, Linear Control Design, Nonlinear Control, Design and Control of Haptic Systems*

Work Experience

FoodTools *07/2024 - 07/2025*
Jr. Mechanical Design Engineer Santa Barbara, CA
- Designed custom parts and assemblies for automated food portioning machines in SolidWorks
- Refined the machining process of intricate parts using Fusion 360 Manufacturing (CAM)
- Drafted and updated engineering drawings in SolidWorks
- Supported the assembly team in wiring and assembling electrical panels, pneumatic control systems, and machinery
- Developed user-facing graphical user interfaces (GUIs) with Python to streamline workflow

Introba *12/2023 - 06/2024*
New Technologies Research Intern Santa Barbara, CA
- Worked in an interdisciplinary team of engineers and environmental equity experts to develop the UCSB Clean Energy Master Plan
- Researched energy storage technologies and HVAC systems for a planned central utility plant
- Developed calculators that modeled the energy needs of buildings around the UCSB campus in Microsoft Excel
- Designed a seawater heat exchange system in SolidWorks

Meissner *06/2023 - 09/2023*
Research and Development Intern Camarillo, CA
- Conducted research on membrane porosity characterization
- Developed a test method to determine porosity of membranes across varying porosities
- Designed and machined laboratory equipment using manual mills and lathes
- Performed dynamic mechanical analysis to determine the effects of solvents on mechanical properties of epoxies
- Completed formal documentation, technical reports, and presentations to share results and data

Projects

Terrain Simulator *01/2026 - 03/2026*
Graduate Student San Diego, CA
- Designed and built a 5-bar kinesthetic lower-limb haptic device for simulating variable terrain underfoot
- Integrated BLDC motors, VESC motor controllers, incremental encoders, and a Teensy 4.1 microcontroller into a complete electromechanical hardware system
- Programmed 5 static and 2 dynamic virtual environments using impedance control in Arduino IDE, delivering real-time force feedback for terrain simulation
- Analyzed joint torques in Python to achieve 15 lb of end effector resistance across the desired workspace
- Designed mechanical components in Fusion 360 and fabricated them via manual milling and 3D printing

Senior Capstone Project *09/2023 - 06/2024*
Undergraduate Student Santa Barbara, CA
- Developed an assistive mobility device for young children with cerebral palsy or other severe motor impairments
- Integrated BLDC motors, an Arduino Mega, and various sensors into the electromechanical system
- Designed and modeled multiple components for the device using SolidWorks
- Utilized finite element analysis (FEA) to evaluate mechanical components of the system under various loads
- Fabricated major mechanical components using techniques including milling, water jetting, and 3D printing
- Received the Distinguished Technical Achievement in Mechanical Engineering award

Certifications and Courses

- **Coding for Everyone: C and C++ Specialization** (UCSC / Coursera) *03/2025 - 05/2025*
- **Machine Learning with Python** (IBM / Coursera) *03/2025 - 04/2025*

Technical Skills

Skills: MATLAB, SolidWorks, CAM, Igor, COMSOL, Python, C/C++, Arduino IDE, Microsoft Office